E-Commerce Architecture

Proposal & Comparison

Clothing Brand Digital Storefront

This document presents every architecture option considered for your e-commerce project — from simple managed solutions to fully custom stacks. Each option is evaluated on cost, traffic capacity, ease of management, and technical complexity.

Prepared by your development team

CONTENTS

What's Inside

ARCHITECTURE 1

WordPress + WooCommerce + Vercel

The classic approach: WordPress manages products and orders; Next.js delivers the storefront.

HOW IT WORKS

```
Browser
■
▼
Vercel (Next.js frontend — global CDN)
■ REST API calls
▼
WordPress + WooCommerce (backend — any PHP host)
■
▼
MySQL Database + Media files on server
```

TRAFFIC CAPACITY

Droplet size	Concurrent users	Monthly visitors	Notes
$12/mo host	~40 users	~20k/month	Too small — MySQL starves PHP
$24/mo host	~200 users	~80k/month	Good for a new brand
$48/mo host	~500 users	~200k/month	Handles campaigns and flash sales

COSTS

Service	Cost/mo	Notes
Vercel (frontend)	Free	Hobby plan covers a new store
Shared PHP hosting	$12–48/mo	Hostinger, SiteGround, WP Engine
Domain	~€15/year	Registered separately
WooCommerce plugins	$0–150/year	Stripe, PDF invoices, etc.
Stripe payment processing	1.5% + €0.25	Per transaction (EU cards)
Total (starting)	~$17–53/mo	Excluding transaction fees

PROS & CONS

Advantages	Limitations
✓ Non-technical staff can manage products and orders immediately	✗ WordPress + plugins slow the API down significantly
✓ 300+ payment plugins — one-click Stripe, PayPal, Klarna, Mollie	✗ Security patches and plugin updates require ongoing maintenance
✓ WooCommerce order dashboard is polished and client-friendly	✗ Limited control — plugin conflicts are common
✓ Huge plugin ecosystem — reviews, loyalty, abandoned cart emails	✗ Performance ceiling lower than a custom API
✓ No custom development for admin features	✗ Not truly headless — some WooCommerce logic leaks to frontend

Verdict: Best for teams that want zero custom code and a polished admin out of the box. Accept a performance and maintenance cost in return.

ARCHITECTURE 2

Vercel + Cloudways Autonomous

WordPress on a Kubernetes-backed managed platform that autoscales automatically.

HOW IT WORKS

```
Browser
■
▼
Vercel (Next.js — global CDN, autoscales)
■ REST API calls
▼
Cloudways Autonomous (Kubernetes cluster)
■■ WordPress + WooCommerce pods [scales 1 → N]
■■ Managed MySQL + Redis cache
```

TRAFFIC CAPACITY

Droplet size	Concurrent users	Monthly visitors	Notes
Entry plan	~300 users	~120k/month	Autoscales within plan limits
Growth plan	~1,000 users	~400k/month	Kubernetes adds pods on spikes
Scale plan	Unlimited	Millions	True horizontal scaling

COSTS

Service	Cost/mo	Notes
Vercel (frontend)	Free	Hobby plan
Cloudways Autonomous	$35–120/mo	Entry to Growth plan
Domain	~€15/year	Registered separately
Stripe processing	1.5% + €0.25	Per transaction
Total	~$35–120/mo	Scales with your traffic

PROS & CONS

Advantages	Limitations
✓ True autoscaling — survives a viral Instagram post	✗ More expensive than a single Droplet at low traffic
✓ Managed infrastructure — no server maintenance	✗ Still carries WordPress performance overhead
✓ WordPress admin is client-friendly out of the box	✗ Less control than a self-managed server
✓ Cloudflare Enterprise CDN built in	✗ Plugin updates still required
✓ Daily backups and one-click restores included	✗ Headless WooCommerce has some compatibility quirks

Verdict: Best choice if you want WordPress's ease of use AND protection against traffic spikes, and don't mind paying a premium for it.

ARCHITECTURE 3

Vercel + DigitalOcean Autoscale Pool

Multiple identical servers behind a load balancer, sharing one database.

HOW IT WORKS

```
Browser
■
▼
Vercel (Next.js — global CDN)
■ REST API calls
▼
DO Load Balancer ($12/mo)
■■ Droplet 1 (WordPress + Nginx) always on
■■ Droplet 2 (WordPress + Nginx) always on
■■ Droplet N (auto-added on spike) per-second billing
■ all share
■■■■■■■■■■■■■■■■■■■■■■
Managed MySQL ($15/mo) DO Spaces ($5/mo)
```

TRAFFIC CAPACITY

Droplet size	Concurrent users	Monthly visitors	Notes
2× $24 Droplets	~400 users	~160k/month	Baseline — always running
4× $24 Droplets	~800 users	~320k/month	Auto-added during spikes
N× Droplets	Linear scale	Unlimited	Pay per second for extra nodes

COSTS

Service	Cost/mo	Notes
Vercel (frontend)	Free	
DO Load Balancer	$12/mo	Always on
2× Base Droplets	$48/mo	WordPress + Nginx
Managed MySQL	$15/mo	Shared by all Droplets
DO Spaces + CDN	$5/mo	Shared media storage

Extra Droplets	Cents/spike	Per-second billing
Total (normal)	~$80/mo	Rises during traffic spikes

PROS & CONS

Advantages	Limitations
✓ True horizontal autoscaling on DigitalOcean	✗ More complex to set up than a single Droplet
✓ No single point of failure — load balancer routes around crashes	✗ Requires shared media storage (DO Spaces) — extra config
✓ Per-second billing makes spike costs minimal	✗ Still carries WordPress overhead on each node
✓ Full control over server configuration	✗ Managed MySQL is a separate cost even at low traffic
✓ Cheaper than Cloudways at the same traffic level	✗ Database becomes bottleneck at very high scale

Verdict: Best for brands that want self-managed infrastructure with real autoscaling. More work to set up; pays off at medium to high traffic.

ARCHITECTURE 4

Vercel + Single DigitalOcean Droplet

Everything on one server: WordPress, MySQL, and Redis. Simple and affordable.

HOW IT WORKS

```
Browser
■
▼
Vercel (Next.js — global CDN)
■ REST API calls
▼
Single DigitalOcean Droplet
■■ Nginx (reverse proxy)
■■ WordPress + WooCommerce (PHP-FPM)
■■ MySQL (local database)
■■ Redis (object cache)
■
DO Spaces (product images + CDN)
```

TRAFFIC CAPACITY (WordPress stack)

Droplet size	Concurrent users	Monthly visitors	Notes
$12/mo 1 vCPU 1GB	~40 users	~20k/mo	Too small for MySQL + PHP together
$24/mo 2 vCPU 4GB	~200 users	~80k/mo	Sweet spot for a new brand
$48/mo 4 vCPU 8GB	~500 users	~200k/mo	Handles campaigns and flash sales
$96/mo 8 vCPU 32GB	~1,500 users	~600k/mo	Single server limit — consider splitting

COSTS

Service	Cost/mo	Notes
Vercel (frontend)	Free	

DO Droplet $24	$24/mo	2 vCPU, 4GB RAM — recommended start
DO Spaces + CDN	$5/mo	Product images
Stripe processing	1.5% + €0.25	Per transaction
Total	~$29/mo	Cheapest production-grade option

PROS & CONS

Advantages	Limitations
✓ Cheapest production setup at ~$29/mo	✗ No autoscaling — a traffic spike can take the site down
✓ Simple — one server to manage	✗ Single point of failure — one crash, everything is offline
✓ Easily resized vertically in the DO dashboard	✗ WordPress performance overhead still applies
✓ Clear upgrade path to autoscale pool when needed	✗ MySQL and PHP compete for the same RAM
✓ Full control over configuration	✗ Must manually resize to handle growth

Verdict: Best starting point for a new brand. Low cost, easy to manage. Upgrade to Architecture 3 when you need autoscaling.

ARCHITECTURE 5

Vercel + AWS (S3 + Lambda + RDS)

Fully serverless: pay only for what you use, scales to millions automatically.

HOW IT WORKS

```
Browser
■
▼
S3 + CloudFront (Next.js static assets — or Vercel)
■ API calls
▼
API Gateway (HTTP API — $1/million requests)
■
▼
Lambda Functions (serverless — pay per execution)
■■ /products /cart /orders /payments
■
▼
RDS MySQL (managed DB — fixed monthly cost)
+ NAT Gateway (Lambda → RDS in VPC) ■ hidden cost
```

TRAFFIC CAPACITY

Droplet size	Concurrent users	Monthly visitors	Notes
Free tier	~1M req/mo	Lambda free tier	No cost for compute at low scale
db.t3.micro	Low-medium	~40k visitors/mo	$22/mo fixed even at zero traffic
db.t3.small	Medium	~200k visitors/mo	$30/mo — grows with product catalog
Unlimited	Infinite	Autoscales with Lambda	DB is the only bottleneck

COSTS

Service	Cost/mo	Notes
S3 + CloudFront	~$5/mo	Frontend assets
API Gateway	~$5–12/mo	HTTP API, pay per request

Lambda compute	~$3–8/mo	Nearly free at small scale
RDS MySQL t3.small	~$30/mo	Fixed — runs 24/7
NAT Gateway	~$33/mo	Required for Lambda → RDS. Hidden cost!
CloudWatch logs	~$5–10/mo	Monitoring
Total	~$90–120/mo	Much higher than expected due to NAT

PROS & CONS

Advantages	**Limitations**
✓ Infinite autoscaling — Lambda handles any traffic spike	✗ NAT Gateway adds ~$33/mo — often exceeds Lambda cost
✓ Pay per request — zero cost at zero traffic (Lambda)	✗ Most complex setup of all options considered
✓ AWS reliability — 99.99% uptime SLA	✗ RDS is a fixed cost even when the store is quiet
✓ Full control over every component	✗ Requires deep AWS knowledge to maintain
✓ Industry-standard — any developer knows this stack	✗ Cold starts add latency on Lambda first requests
	✗ Overkill for a single clothing brand store

Verdict: Powerful but over-engineered for this project. The NAT Gateway alone costs more than our entire recommended stack. Only choose this if your team already runs AWS infrastructure.

ARCHITECTURE 6

Vercel + Django API + DigitalOcean

Custom Python API replaces WordPress entirely. Maximum performance, full control.

HOW IT WORKS

```
Browser
■
▼
Vercel (Next.js — global CDN, autoscales free)
■ REST API calls
▼
DigitalOcean Droplet ($24/mo)
■■ Nginx (reverse proxy + SSL + rate limiting)
■■ Django + Gunicorn (custom REST API)
■■ MySQL (products, orders, customers)
■■ Redis (cart sessions, product cache)
■
DO Spaces + CDN (product images — $5/mo)
■
Stripe (payments — no card data touches our server)
```

TRAFFIC CAPACITY (Django stack — 2–3× more than WordPress)

Droplet size	Concurrent users	Monthly visitors	Notes
$12/mo 1 vCPU 1GB	~100 users	~40k/mo	Minimum viable — dev/staging only
$24/mo 2 vCPU 4GB	~500 users	~200k/mo	Recommended start — sweet spot
$48/mo 4 vCPU 8GB	~1,200 users	~500k/mo	Growing brand — handles flash sales
$72/mo 8 vCPU 16GB	~2,500 users	~1M+/mo	Consider separate DB at this level

WHY DJANGO HANDLES MORE TRAFFIC ON SAME HARDWARE

	Django (this stack)	WordPress (same Droplet)

Idle RAM usage	~180 MB	~380 MB
Code on each request	Only your code runs	10–30 plugins execute
Product page with Redis	0 DB queries (cached)	3–8 DB queries
Checkout (uncacheable)	Same — hits DB live	Same — hits DB live

COSTS

Service	Cost/mo	Notes
Vercel (frontend)	Free	Hobby plan covers any new store
DO Droplet $24	$24/mo	Django + MySQL + Redis + Nginx
DO Spaces + CDN	$5/mo	250GB product images + global CDN
Domain	~€15/year	.gr domain
dev.domain.gr	$12/mo	Separate dev Droplet — optional
Stripe processing	1.5% + €0.25	Per transaction (EU cards)
Total (production)	~$29/mo	Same cost as WordPress — 2.5× faster

WHAT THE CLIENT MANAGES DAY-TO-DAY

Task	How	Difficulty
Add a product (with sizes/colours)	Django admin — inline variant form	Easy after setup
Process an order	Change status → auto-emails customer	Very easy
Run a discount / sale	Create discount code in admin	Easy
Upload product images	Drag and drop in admin	Easy
View sales reports	Custom dashboard page in admin	Read-only — very easy
Issue a refund	Stripe dashboard (separate tab)	Easy
Bulk upload products via Excel	django-import-export plugin	Easy — one upload

PROS & CONS

Advantages	Limitations
✓ 2–3× more traffic than WordPress on identical hardware	✗ Requires 1–2 days of custom admin configuration
✓ No plugin tax — only your code runs on every request	✗ Admin is not as polished as WooCommerce out of the box
✓ Full control — any feature can be built exactly as needed	✗ Refunds go through Stripe dashboard separately
✓ Redis caches product pages — near-zero DB load when browsing	✗ Any new admin feature needs development time
✓ Custom admin can be tailored to the client's exact workflow	✗ More initial setup work than WordPress
✓ Same $29/mo cost as the cheapest WordPress setup	
✓ Clear upgrade path — add autoscale pool when needed	

Verdict: OUR RECOMMENDATION. Same monthly cost as WordPress, 2.5× the traffic capacity, and a cleaner codebase. 1–2 days of admin customisation gives the client a comfortable, tailored experience. The right choice for a brand that wants to grow.

SUMMARY

All Architectures at a Glance

Use this table to compare every option side by side.

WordPress + Vercel (shared host)	$17–53	80k–200k/mo	No	★★★★★	Low	Starting out
Cloudways Autonomous + Vercel	$35–120	400k+/mo	Yes	★★★★★	Low	Easy + scale
DO Autoscale Pool + Vercel	$80+	Unlimited	Yes	★★★	Medium	Medium traffic
DO Single Droplet + WP + Vercel	$29	80k–200k/mo	No	★★★★★	Low	Cheapest start
AWS Lambda + RDS + Vercel	$90–120	Unlimited	Yes	★★	High	AWS teams only
Django + DO Droplet + Vercel ❖	$29	200k–500k/mo	No→Yes*	★★★★	Medium	OUR PICK

* Django stack on single Droplet has no autoscale. Adding the DO autoscale pool later is a straightforward migration.

RECOMMENDATION

What We Recommend and Why

After evaluating all six architectures against your requirements — a clothing brand e-commerce site with non-technical staff, low admin overhead, good performance, and room to grow — we recommend:

Architecture 6	**Vercel + Django API + DigitalOcean Droplet**

- **Same price as the cheapest option**
  $29/mo — identical to a basic WordPress setup, but with 2.5× the traffic capacity.
- **Best performance per euro**
  Django's idle memory is half of WordPress. Redis caches product pages so the database is barely touched during browsing. Your storefront will load faster.
- **Client-friendly admin**
  1–2 days of customisation gives the brand team a tailored dashboard: inline product variants, order status with auto-emails, discount codes, sales reports, and bulk Excel product upload.
- **No plugin anxiety**
  No WordPress plugins to update, no security patches from third-party authors, no plugin conflicts. Only your code runs.
- **Clear upgrade path**
  When traffic grows, moving MySQL to a Managed Database costs $15/mo extra. Adding the autoscale pool is a straightforward migration with no downtime.
- **Production-grade security built in**
  JWT authentication, Stripe webhook signature verification, atomic stock decrement, rate limiting, HTTPS, and admin 2FA are all included.

SCALING ROADMAP

Launch	$29/mo	Single Droplet + Django + Vercel. Handles up to 500 concurrent users.
Growth	$44/mo	Add Managed MySQL (+$15/mo). Frees Droplet RAM for more traffic.
Scale	$80+/mo	Add DO Load Balancer + autoscale pool. Handles any traffic spike.

Enterprise	Custom	Multi-region deployment, read replicas, dedicated CDN contracts.